

(SoM) solutions. These modules are centred around Xilinx FPGAs and micro-semi PolarFire RISC V FPGAs. One of our notable products in this category is the AMD Zynq UltraScale+ MPSoC, which is a multi-processor system on a chip. This advanced processor features a Quad-core ARM Cortex-A53, Dual-core ARM Cortex-R5, Mali GPUs, and supports H.264 and H.265 video codecs. Furthermore, it boasts an FPGA logic cell capacity that can scale up to a quarter million cells.

Are the SoMs completely designed and manufactured by your teams or just assembled?

Yes, our system-on-module (SoM) products are entirely designed by our in-house team. This includes not just the layout but also the assembly and BGAs (ball grid arrays). We focus on building a complete system model around the chipset rather than engaging in chip-level design. We develop the entire board as a system model. The standards for SoM are primarily set by a German consortium known as SgET, of which we are a member. We use their specifications as a guideline to design our system on modules.

Till now, these SoM form factors have been tailored for traditional ARM processors, among others. FPGA modules, however, tend to be more customisable based on specific customer requirements due to the inherent flexibility of FPGAs. By April next year, SgET is expected to release a universal form factor for FPGAs too, which should further standardise and streamline the design process for these components.

Where is the R&D and manufacturing of your boards conducted?

The research and development (R&D) of our boards is entirely conducted in India. All engineering and manufacturing processes are based in India too. However, we outsource the fabrication of PCBs (printed

circuit boards) predominantly to Taiwan. This decision is not due to a lack of capability in India, but rather due to concerns about yield rates and lead times. Taiwanese suppliers are able to fabricate and ship PCBs more quickly and efficiently than their counterparts in India. In addition to PCB fabrication, every component is sourced by Numato Lab itself.

Are you open to other sectors like automotive and robotics or limited to drones only?

We are certainly open to collaboration in these areas. We are gradually expanding into the automotive sector. Currently, we have clients such as Ford, Tesla, who purchase our relay modules for use in their manufacturing plants. Our focus remains primarily on industrial applications and our products are relevant in the context of Industry 4.0. If a customer approaches us, we are fully prepared to work with them. We are even open to pitching. We are planning to conduct a series of webinars with Microchip and Future to showcase our capabilities. Our platform is versatile and designed for a broad spectrum of applications. I highlighted the focus on drone customers because we already have a partnership in this sector. Moreover, one of our products, the Spartan 6 module, is currently being used by a Swiss medical equipment company. They employ it in endoscopy applications, which require handling multiple camera inputs and controlling the stepper motor of the endoscope. This process is executed using our FPGA.

Do you have to get medical grade certification in such cases or do customers get it themselves?

The customers are responsible for obtaining the medical grade certification. As a system, they will ensure that the necessary certifications are in place. From our end,



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